LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	28
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Maintenance of O2 system, servicing and safety precautions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Maintenance of O2 system, servicing and safety precautions. Safety measures while handling /upkeep of gaseous oxygen system. Quality checks on pressurized gas cylinders.
5. Specific Objectives	17.2.1.1 Trainee will be able to understand the need of oxygen systems, types of systems, working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of O2 cylinder quality checks and safety valve inspection; Instructor supervision of trainee handling procedures.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft oxygen system?
(b) List two critical safety precautions when handling gaseous oxygen.
(c) Why must oil or grease be strictly avoided on oxygen equipment?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Need and types of oxygen systems • Working principles of O2 components • Safety measures for gaseous oxygen handling • Quality checks on pressurized cylinders	L & Ds	PPT, IB	15 min
(b)	• Servicing procedures for O2 cylinders • Inspection of safety valves and regulators • Leak detection techniques and protocols • Storage and transportation safety rules	L & Ds	PPT, PDF	20 min
(c)	• Review of maintenance logs and records • Common defects and troubleshooting steps • Final safety checklist for O2 systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct procedure for checking the hydrostatic test date on an O2 cylinder.
(b) Describe the immediate actions required if a leak is detected in the cabin O2 system.

12. Home Assignment and Follow Up	Prepare a brief report on the specific safety precautions required when servicing gaseous oxygen cylinders, referencing the next lesson on cabin pressurization systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Typical cabin pressurization system, Cabin pressure regulator safety v/v, outflow v/v etc. and its operation, Methods of sealing of cabin	Refer to the master syllabus reference list	LP 29

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min